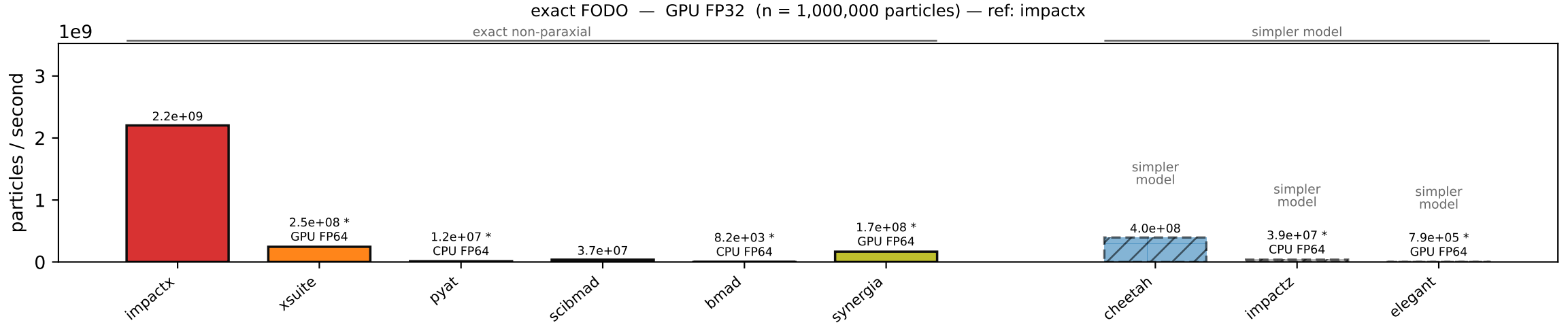


*'simpler model' = a lower-fidelity model vs the reference — cheetah: paraxial chromatic-linear quad; impactz: paraxial chromatic thick quad (transfmapK, K/(1+delta)); elegant: trace-space slopes + achromatic drift*

** next-best shown: xsuite, synergia, elegant = GPU FP64 (no FP32); pyat, bmad, impactz = CPU FP64 (no GPU)*

versions: impactx 26.08 · xsuite 0.53.4 · pyat 0.8.0 · scibmad 0.4.1 · bmad 20260904-1 · synergia v2024.04-209-g795b99f39 · cheetah 0.8.2 · impactz V2.7.7 · elegant elegant-2026.4.0